

assignment6 - PE and KE , intro to capacitor ↕

⚠ This is a preview of the published version of the quiz

Started: Mar 22 at 7:10pm

Quiz Instructions

I will post a tutorial for the questions with Math



Question 1 9 pts

How much kinetic energy does a proton gain if it is accelerated, with no friction, through a potential difference of 1.00 V? The proton is 1836 times heavier than an electron, and $e = 1.60 \times 10^{-19} \text{ C}$.

☐

$1.60 \times 10^{-19} \text{ eV}$

☐

1836 J

☐

1836 eV

☐

1.00 eV

☐

1.00 J



Question 2 9 pts

A tiny particle with charge $+ 5.0 \text{ } \mu\text{C}$ is initially moving at 55 m/s. It is then accelerated through a potential difference of 500 V. How much kinetic energy does this particle gain during the period of acceleration?

☐

$2.5 \times 10^{-3} \text{ J}$

☐

$1.0 \times 10^4 \text{ J}$

☐

100 J

☐

2500 J



Question 3 9 pts

A proton that is initially at rest is accelerated through an electric potential difference of magnitude 500 V.

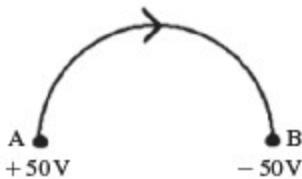
What speed does the proton gain? ($e = 1.60 \times 10^{-19} \text{ C}$, $m_{\text{proton}} = 1.67 \times 10^{-27} \text{ kg}$)

☐ $2.2 \times 10^5 \text{ m/s}$ ☐ $3.1 \times 10^5 \text{ m/s}$ ☐ $1.1 \times 10^5 \text{ m/s}$ ☐ $9.6 \times 10^5 \text{ m/s}$

⋮

Question 4 9 pts

How much work must we do on an electron to move it from point A, which is at a potential of +50V, to point B, which is at a potential of -50 V, along the semicircular path shown in the figure? Assume the system is isolated from outside forces. ($e = 1.60 \times 10^{-19} \text{ C}$)

☐

-1.6 J

☐

1.6 J

☐

This cannot be determined because we do not know the distance traveled.

☐ $1.60 \times 10^{-17} \text{ J}$ ☐ $-1.60 \times 10^{-17} \text{ J}$

⋮

Question 5 9 pts

After a proton with an initial speed of $1.50 \times 10^5 \text{ m/s}$ has increased its speed by accelerating through a potential difference of 0.100 kV, what is its final speed? ($e = 1.60 \times 10^{-19} \text{ C}$, $m_{\text{proton}} = 1.67 \times 10^{-27} \text{ kg}$)

☐ $8.80 \times 10^5 \text{ m/s}$ ☐ $1.55 \times 10^6 \text{ m/s}$

 $4.56 \times 10^5 \text{ m/s}$  $3.55 \times 10^5 \text{ m/s}$  $2.04 \times 10^5 \text{ m/s}$ 

Question 6 9 pts

A proton with a speed of $2.0 \times 10^5 \text{ m/s}$ accelerates through a potential difference and thereby increases its speed to $4.0 \times 10^5 \text{ m/s}$. Through what magnitude potential difference did the proton accelerate? ,

$m_{\text{proton}} = 1.67 \times 10^{-27} \text{ kg}$)



840 V



1000 V



630 V



100 V



210 V



Question 7 9 pts

A proton that is initially at rest is accelerated through an electric potential difference of magnitude 500 V.

How much kinetic energy does it gain? ($e = 1.60 \times 10^{-19} \text{ C}$)

 $8.0 \times 10^{-17} \text{ J}$ 

800 J

 $1.6 \times 10^{-19} \text{ J}$ 

500 J



Question 8 9 pts

https://phet.colorado.edu/sims/html/capacitor-lab-basics/latest/capacitor-lab-basics_en.html

(https://phet.colorado.edu/sims/html/capacitor-lab-basics/latest/capacitor-lab-basics_en.html)

Open this App in your browser. Click on CAPACITANCE.

check ALL the boxes. ALL

Observe the gauges (capacitance, charge, energy) and the simulation

Note: the capacitance is the amount of charge a capacitor can hold per unit voltage.

Increase the voltage of the battery toward 1.5V. What is true as you increase the voltage of the battery

- ☐ the energy increases, the charge and the capacitance stay constant.
- ☐ the charge increases, the energy increases, the capacitance stays constant
- ☐ nothing change
- ☐ everything increases



Question 9 9 pts

https://phet.colorado.edu/sims/html/capacitor-lab-basics/latest/capacitor-lab-basics_en.html 
(https://phet.colorado.edu/sims/html/capacitor-lab-basics/latest/capacitor-lab-basics_en.html)

Building on previous question.

Leave the power supply at 1.5V.

The area of the plate is 200 mmxmm, Increase the area to 400 mmxmm

Take note of the numbers next to the gauges before and after the change.

What is true (about)

- ☐ capacitance decreases by a factor of 2
- ☐ capacitance increases by a factor of 2
- ☐ capacitance stays constant
- ☐ capacitance increases by a factor of 4



Question 10 9 pts

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(https://phet.colorado.edu/sims/html/capacitor-lab-basics/latest/capacitor-lab-basics_en.html)

Building on previous question.

Leave the power supply at 1.5V.

The area of the plate is 200 mmxmm, Increase the area to 400 mmxmm

Take note of the numbers next to the gauges before and after the change.

What is true (about)

☐

Energy and charge stay constant

☐

energy and charge decrease by a factor of 2

☐

Energy and charge increase by a factor of 2

☐

energy and charge increase by a factor of 3



Question 11 9 pts

https://phet.colorado.edu/sims/html/capacitor-lab-basics/latest/capacitor-lab-basics_en.html 
(https://phet.colorado.edu/sims/html/capacitor-lab-basics/latest/capacitor-lab-basics_en.html)

Building on previous question.

Leave the power supply at 1.5V.

Decrease the distance between the 2 plates

Take note of the numbers next to the gauges before and after the change.

What is true (about)

☐

the charge stays constant

☐

The charge increases

☐

IDK

☐

the charge decreases



Question 12 9 pts

https://phet.colorado.edu/sims/html/capacitor-lab-basics/latest/capacitor-lab-basics_en.html 
(https://phet.colorado.edu/sims/html/capacitor-lab-basics/latest/capacitor-lab-basics_en.htm)

refresh and now click on light bulb. Increase the voltage to 1.5V. The capacitor is charged. Disconnect the capacitor and connect it to the light bulb. Whats happening

☐

the light bulb keeps the same brightness as the capacitor loses its charge

☐

absolutely nothing happen

☐

the light bulb stays off and the capacitor stays charged

☐

The light bulb is very bright then dimmer. The capacitor loses its charge

Not saved

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